Historical source: CHEMISTRY

continued from the previous page

[Margin: Magnella, Sc. | 334 Prepris-tion. | 1553 Properties | 1354 Achon of air, Sc. | 1355 Prepras-tion. | 1359 Properties | 1357 Achon: heat. | 1358 History and preparation. | 1358 History and preparation. | 1359 Properties | 1356 Of heat. | 1354 Competi-tion.]

3. Sulphite of ~~Magnella.~~ **Magnesia.** 1. The compound of fulphurous acid and magnella is formed by pataling fulphurous acid gas into two parts of water, with one of carbonate of magnella. A violent effervescent takes place, with the evolution of heat. The fulphite of magnella is formed, and precipitated to the bottom in the fate of powder / but with an excess of acid it is reaffilatively, and crytallizes. 2. The ~~crythals~~ **cryftals** of ~~fulphirous magnella~~ **fulphite of magnesia** are in the form of depredited transparent tetrahedrons. It has a mild earthy tafe, which ~~then~~ **foon** becomes ~~family fulphra round;~~ **fenibly fulphurous;** it has no final. Its ~~(pecie~~ **fpecific** gravity is 1) 3/5oz. 3. It ~~effectes~~ **effloresces** in the air, and is ~~flowed~~ **slowly** converted into ~~fulphate~~ **sulphate** of ~~magenae.~~ **magnesia.** It is ~~foluble~~ **soluble** in 20 parts of cold water. Boiling water ~~didties~~ **dissolves** a greater proportion, and from this ~~fit~~ **it** crystallizes on cooling. Exposed to heat, this ~~felt~~ **salt** becomes ~~vicidé,~~ **viscous,** and by calcination it ~~lofea 0.4;~~ **loses 0.45** of its weight. If the heat be increased, it is decomposed, the acid is driven off, and the pure earth remains behind. 3. It effectes in the air, and is flowed converted into fulphate of magenae. It is soluble in 20 parts of cold water. Boiling water didtics a greater proportion, and from this fit crystallizes on cooling. Exposed to heat, this felt becomes vicidé, and by calcination it lofea 0.4; of its weight. If the heat be increased, it is decomposed, the acid is driven off, and the pure earth remains behind. The component parts of this fall are, The component parts of this fall are, 1. This triple ~~falls~~ **falt** is formed by decomposing the fulphite of ammonia by ~~magnella~~ **magnesia,** or the fulphite of ~~magenae~~ **magnesia** by ~~ammonium. In foliation~~ **ammonia, in solution** in the ~~cold /~~ **cold;** or, by mixing together the ~~folutions~~ **solutions** of the two ~~falls.~~ **falts.** 2. This ~~falls~~ **fult** is in ~~nonfrentacyrdia,~~ **tranfparent cryftals,** the form of which has ~~had~~ **not** been determined. When it is ~~propelled~~ **expofed** to the air, it is converted into ~~fulphate~~ **fulpbate** of ammonia and ~~magenails.~~ **magnesia.** It is ~~lecf fulphin~~ **lefs fuble** in water than either of the two ~~fulphite~~ **fulpbites** of which it is formed. By the action of heat, fulphurous acid is given ~~on,~~ ~~aculusful fulphite~~ **out, acidulous fulpbite** of ammonia is fublimed, and there remains behind pure 5. Nitrate of ~~Magnella.~~ **Magnesia.** 1. This compound of nitric acid and magnella was formerly called _nitre with belfs of magnella_, and magenina _lolpetere_. It is formed by the direct combination of the acid with the earth. By evaporation it is crytallized. 2. This ~~fatty crytlinalizes~~ **falt cryftallizes** in ~~four-filled~~ **fourfided** rhomboidal ~~pyrite,~~ **prifms,** whole ~~fuminis~~ **fummits** are oblique or truncated. ~~Some times~~ **Sometimes** it is in the form of ~~small~~ **fmall** needles combined in groups. The tafte is penetrating and bitter. The ~~pecitie~~ **fpecific** gravity is ~~1.75.~~ **1.736.** 3. It is ~~deliquent~~ **deliquescent** in the air, and is ~~fsoluble~~ **soluble** in its own weight of cold water. It is more ~~foluble~~ **soluble** in boiling water, in which it ~~crytallizes~~ **crystallizes** on ~~cooling ; but,~~ **cooling; but** it can only be obtained in regular crystals by ~~flow~~ **slow** evaporation from its ~~foliation~~ **solution** in cold water. 1360 Action of water. 4. By the action of heat it undergoes the ~~waters fufen,~~ **watery fufion;** the water is driven off, and it becomes dry. It is ~~decompiled~~ **decomposed** in a ~~throat,~~ **strong heat,** gives out a little oxygen gas, then nitrous gas, and at ~~all~~ **last** the nitric acid. The pure earth remains behind. The component parts of this fall are, according to

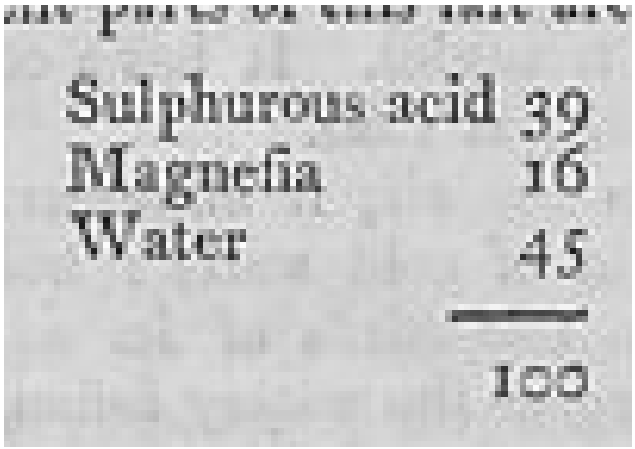


Plate: Sulphurous acid 39 Magnella 16 Water 45

CHEMISTRY

CH E M I S T R Y. \begin{tabular}{cccc} \thead{\mbox{}} & \thead{Bergman.} & \thead{Kirwan.} & \thead{Magne}\\ Acid & 43 & 46 & \mbox{}\\ Magnella & 27 & 22 & \mbox{}\\ Water & 30 & 32 & \mbox{}\\ 100 & 100 & 100 & \\ \end{tabular} 6. Nitrate of Ammonia and ~~Magnella.~~ **Magnesia.** 1. This triple ~~falls~~ **falt** is formed, either by the ~~direcliorgan~~ **direct** combination of the ~~folutions~~ **solutions** of nitrate of ammonia, and nitrate of ~~magnella,~~ **magnesia,** by which the ~~falls~~ **falt** is obtained pure and ~~crytallized ,~~ **cryftallized ;** or, by partially decomposing the nitrate of ammonia by ~~matguela,~~ **magnesia,** or the nitrate of ~~magenia~~ **magnesia** by ammonia. 2. The crythals of this fall are and the form of free frozen grains, it has a bitter, acids, and ammonated tafte. It is lefs deliquent in the air than either of the confluent fats, and lefs foluble in water. It requires 11 parts of cold water as differs it, but lefs of boiling water. It crytallizes on cooling. When it is rapidly heated, it ~~inflamed spontaneously. After when lonly~~ **inflames fpontaneously. When flowly** heated in ~~cloke vells,~~ **clofe veffels,** it gives out ~~very heat, gen,~~ **oxy-gean**

gas, ~~and~~ **azotic** gas, a greater proportion of water than it contains, nitrous gas, and nitric acid, without the ~~familiar~~ **fmall left** trace of ~~ammonia,~~ **ammonia;** which flows that it is ~~de-completed,~~ **de-compofed,** that the hydrogen combines with the oxygen of the acid, and forms water. The component parts of this fall are, 7. Nitrite of ~~Magnella.~~ **Magnesia.** Nothing is known of the properties of this fall. 8. Muriate of ~~Magnella.~~ **Magnesia.** 1. This compound of ~~miruric~~ **muriatic** acid and ~~magnella~~ **magnesia** found was formerly called ~~mirice, left~~ **marine salt** of ~~magnella~~, **magnesia,** and was not confounded with the muriate of lime, with which it is frequently ~~recommended.~~ **accompanied.** The difference between ~~thefe~~ **these** two ~~falls~~ **salts** was ~~flat printed and~~ **first pointed out** by Dr ~~Blacks,~~ **Black,** and Bergman afterwards examined the nature and ~~proper ties~~ **properties** of muriate of ~~magnella.~~ **magnesia.** The ~~falls~~ **salt** is obtained by ~~differing magnella~~ **dissolving magnesia** in ~~muricate~~ **muriatic** acid till they are ~~fatuated,~~ **saturated,** and then evaporating the solution. Small irregular crystals are obtained. This ~~falls~~ **salt** exists in the waters of the ocean, and in mineral waters, along with the ~~marinate~~ **muriates** of ~~folds~~ **soda** and lime. 2. It is extremely difficult to obtain the muriate of proper magnetic in any regular form. It is either in the late or powder, or very small regular needles or in a kind of jelly. It has a diligenceable bitter tafte. The specific gravity is 1.601. 3. It is very ~~deliquent~~ **deliquefcent** in the air. Cold water readily ~~dilduces~~ **difolves** its own weight, and it is ~~fill~~ **ftill** more ~~lo-ble~~ **foleuble** in boiling water. 4. It is completely ~~decoupled~~ **decomposed** by heat ; the acid is ~~often~~ driven off, and the pure earth remains behind ~~.*~~ ***.** Let it completely downed by heat ; the acid is often driven off, and the pure earth remains behind .* \begin{tabular}{cc} Acid & 34-19\\ Magnella & 41\\ Water & 25\\ \end{tabular}

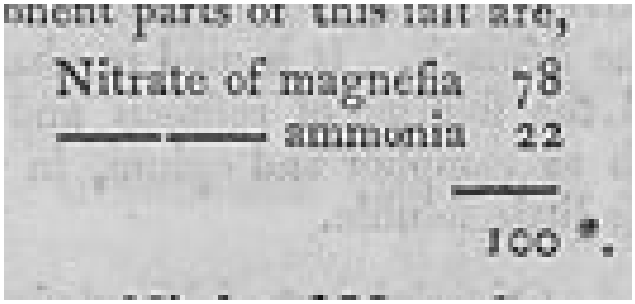


Plate: Nitrate of magnella ammonia 22 100 *

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